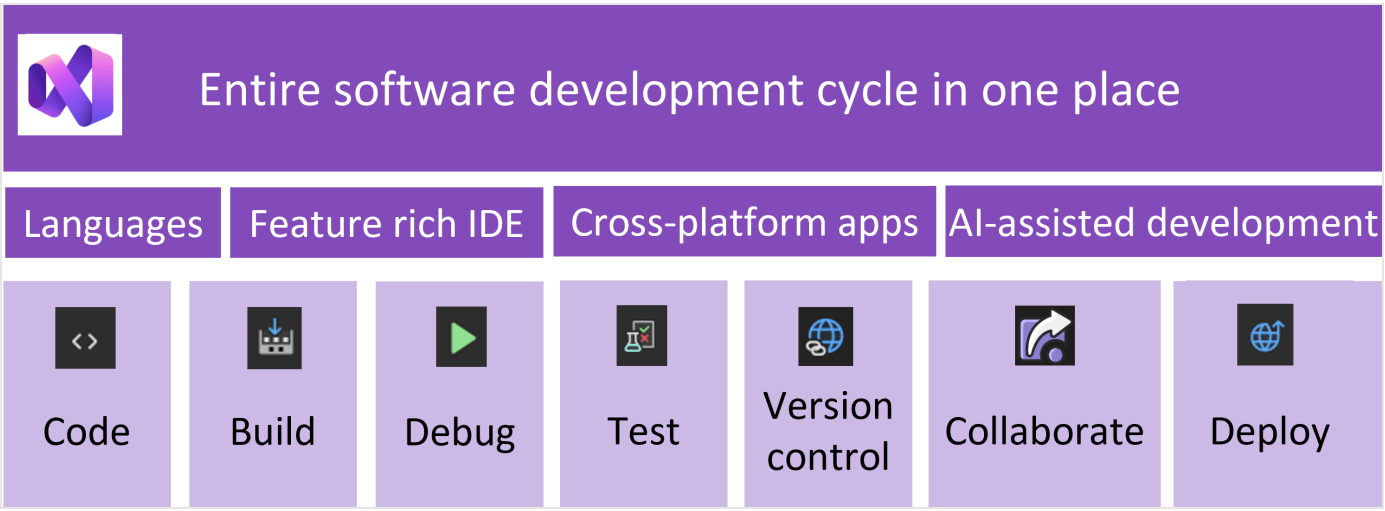
What is Visual Studio?

Visual Studio is a powerful integrated development environment (IDE) for Windows where you can develop, build, debug, test, and deploy your apps, all in one place. With built-in AI assistance from GitHub Copilot, you can accelerate your workflow with intelligent code completions, chat-based guidance, and AI-powered debugging.



Visual Studio includes compilers, code completion tools, source control, extensions, and many more features to enhance every stage of the software development process.

> **Note**
>
> Visual Studio is available for Windows. For a lightweight, cross-platform code editor, see Visual Studio Code .

Install Visual Studio

Visual Studio is available in three editions:

- Community : Free and fully featured for students, open-source developers, and individual developers.
- Professional : For individual developers or small teams.
- Enterprise : For small-to-large business and enterprise organizations.

Compare features across editions to find the edition that best fits your needs.

Download Visual Studio

For more information, see Install Visual Studio.

Why use Visual Studio?

Whether you're writing your first "Hello World" program or deploying enterprise apps, Visual Studio provides the tools you need to work efficiently and collaboratively.

- Multiple-language support. Code in C#, C++, Python, JavaScript, TypeScript, and more.
- Cross-platform development. Build mobile and desktop apps with .NET MAUI, web apps with ASP.NET, or Node.js services.
- AI-assisted development. Write, debug, and test code faster with GitHub Copilot.
- Workload-based installer. Install only the components you need.
- Integrated version control. Clone, branch, commit, and review pull requests by using built-in Git tooling.
- Extensible platform. Customize your IDE with extensions from the Visual Studio Marketplace , or build your own.

Discover Visual Studio

Visual Studio supports different parts of the software development cycle.

Develop your code

Visual Studio Code helps you write and manage code with confidence. Use light bulb suggestions for quick fixes, outlining to collapse code blocks, and **Solution Explorer** or **Class View** to navigate your projects.

Learn more about features in the IDE that help you organize and edit content:

- Code editor
- Tips and tricks

Develop with AI

GitHub Copilot and IntelliCode assist you in writing code faster and with greater accuracy. Use Copilot Completions for inline code suggestions as you type, and Copilot Chat to ask questions, generate code, or refactor existing code through natural language. These tools also help with other development tasks such as writing unit tests, debugging, and profiling.

Learn more about AI-assisted development in Visual Studio.

Build your app

You can compile and build your applications to create builds right away and test them in a debugger. You can run multiprocessor builds for C++ and C# projects. Visual Studio also provides several options that you can configure when you build applications. You can create a custom build configuration, use the built-in configurations, hide certain warning messages, or increase build output information.

Learn more about how to compile and build in Visual Studio.

Debug your code

By using integrated debugging in Visual Studio, you can easily debug, profile, and diagnose code. You can step through your code and look at the values stored in variables, set watches on variables to see when values change, and examine the execution path of your code. Visual Studio also provides other ways to debug your code while it runs.

Learn more about debugging in Visual Studio:

- First look at the debugger
- Debug with Copilot

Test your code

You can write high-quality code by using comprehensive testing tools in Visual Studio. Unit tests give developers and testers a quick way to find logic errors in code. You can analyze how much code you're testing and see instant results in a test suite. Know the impact of every change you make by using advanced features that test code while you type.

Learn more about the testing tools available in Visual Studio:

- Use testing tools in Visual Studio
- Get started with unit testing

Collaborate and manage version control

By using the integrated Git features in Visual Studio, you can clone, create, or open your own repositories. The Git tool window provides everything you need to commit and push changes, manage branches, and resolve merge conflicts. If you have a GitHub account, you can manage repos and review pull requests directly within Visual Studio.

Learn more about version control with Git in Visual Studio.

Deploy your app

When you deploy an application, service, or component, you distribute it for installation on other computers, devices, or servers, or in the cloud. You can choose the appropriate method in Visual Studio for the type of deployment that you need. Share your apps and code by publishing to the web or Azure, or by deploying to a network share or a local folder.

To learn more, see deploy your app from Visual Studio.

Get started

Choose a language-specific tutorial to start building your first app:

- C# — Build a simple console app and learn the basics of C# and .NET.
- C++ — Create a C++ console app and explore the build toolchain.
- Python — Set up Python support and write your first script.
- JavaScript/Node.js — Build a Node.js and Express web app.
- Visual Basic — Create a simple console app with Visual Basic.
- F# — Build a web app with F# and ASP.NET Core.

Next step

- Tour the IDE to get familiar with the IDE features and to learn how to use it for basic tasks.

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